Memory map and peripheral register definitions

Data address = 8-bit value of the address register Rs of LD/ST (ALU output in EX, used in MEM). All writes happen once, in MEM, only for a **valid** instruction (EX/MEM.V & memWrite & region select). Reads are combinational in MEM (one cycle, no READY protocol). Bubbles and flushed instructions can never change RAM or any output.

Range	Region	Read	Write
`0x00 - 0x7F`	Data RAM, 128 x 8 (uninitialised after reset: programs initialise before use)	RAM data	RAM data
`0x80 - 0xEF`	Reserved	0	ignored
`0xF0 - 0xFF`	Peripheral space (decode: addr[7:4] = 1111, index = addr[3:0])	see below	see below

Peripheral registers

Addr	Name	Access	Reset	Bit fields / behaviour
`0xF0`	`IN_CTRL`	R	live	b0 START b2:1 MODE[1:0] b3 DOOR_CLOSED b4 TRIP (read-only status of the safety latch) b7:5 = 0. START, MODE, DOOR are sampled by the LD in MEM; MODE is snapshotted by software on START acceptance.
`0xF1`	`WATER_LEVEL`	R	live	b7:0 unsigned water level (compare with SLTU).
`0xF2`	`TEMP`	R	live	b7:0 temperature readback (sensor only).
`0xF3`	`DRUM_RPM`	R	live	b7:0 drum speed readback (sensor only).
`0xF4`	`ACT`	R/W	0x00	b0 VALVE_IN, b1 HEATER, b2 MOTOR_EN (CPU request), b3 MOTOR_DIR, b4 PUMP_OUT, b5 DOOR_LOCK, b6 BUZZER, b7 reserved (write ignored, reads 0). Physical motor output = b2 AND DOOR_CLOSED AND NOT TRIP.
`0xF5`	`DISPLAY`	R/W	0x00	b7:0 status code / LED pattern.
`0xF6`	`T_COUNT`	R	0x00	b7:0 current count. Writes ignored.
`0xF7`	`T_LIMIT`	R/W	0x00	b7:0 compare value; 0 disables comparison. Writing it never sets MATCH by itself.
`0xF8`	`T_CTRL`	R/W	0x00	b0 EN (reads back as written). b1 CLR: write-1 command, **always reads 0**. b7:2 reserved (ignored, read 0). Write 0x03 = clear + enable (rearm); 0x01 = enable only; 0x00 = stop (COUNT holds).
`0xF9`	`T_STATUS`	R	0x00	b0 MATCH (sticky), b7:1 = 0. Writes ignored.
`0xFA-0xFF`	`(unimplemented)`	R	-	Reads return 0; writes ignored.

Behavioural rules

• Writes to read-only registers (0xF0-0xF3, 0xF6, 0xF9), reserved addresses and unimplemented peripheral addresses are ignored and generate no strobe (observable probes WR_ACT / WR_DISP / WR_TLIM / WR_TCTL stay low; see validation group 3).

• Reset (asynchronous clear of all state registers): ACT = 0 (all actuator requests off), DISPLAY = 0, timer EN = 0, COUNT = LIMIT = MATCH = 0, safety TRIP = 0. RAM and R0-R7 are unspecified; the control program initialises everything it reads.

• Inputs are sampled during the MEM stage of a valid LD; tests hold inputs steady across the sampling clock edge.

Timer control behaviour (T_COUNT / T_LIMIT / T_CTRL / T_STATUS)

Event at a rising clock edge	COUNT	MATCH
Write of T_CTRL with CLR = 1 (priority over everything else)	0	0
EN = 1 and no CLR	COUNT + 1 (mod 256)	set (sticky) if new COUNT = LIMIT and LIMIT != 0
EN = 0 and no CLR	hold	hold
Write of T_LIMIT	unchanged	unchanged (compares the old LIMIT at that same edge)

The timer is clocked by the global clock only, so it keeps counting during load-use stalls, flushes and after HALT. The EN flag is updated by the write itself: writing 0x03 clears at that edge and counting starts at the next edge. Rearm for a new interval: write T_LIMIT, then write T_CTRL = 0x03.

Safety-related readback

IN_CTRL b4 returns the TRIP latch. TRIP is set by hardware when the door is sampled open while ACT b2 is 1; software may only observe it. It clears only when ACT b2 = 0 and a new START low-to-high edge occurs while DOOR_CLOSED = 1.